

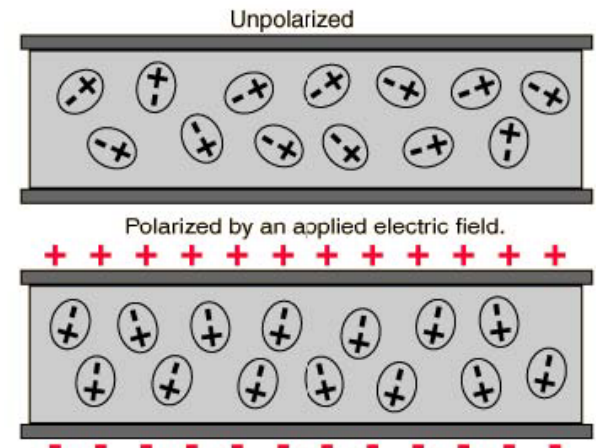
Dispersion

- Dispersion corresponding to the phenomenon whereby the index of refraction of a medium is frequency dependent.

$$n = n(\omega)$$

- Electric polarization, P

$$(\epsilon - \epsilon_0)\vec{E} = \vec{P}$$

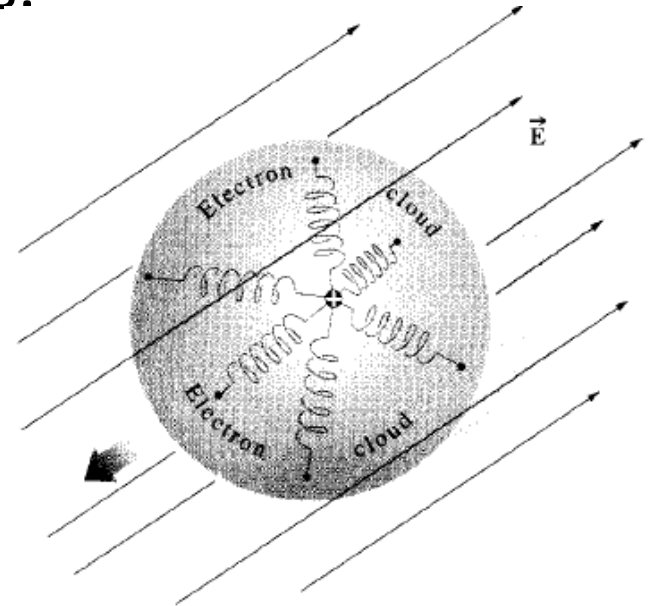


□ If the dielectric is subjected to an incident harmonic electromagnetic wave, its internal charge structure will experience time-varying forces.

□ Classical forced oscillator:

$$q_e E_0 \cos \omega t - m_e \omega_0^2 x = m_e \frac{d^2 x}{dt^2}$$

$$x(t) = \frac{q_e/m_e}{(\omega_0^2 - \omega^2)} E_0 \cos \omega t = \frac{q_e/m_e}{(\omega_0^2 - \omega^2)} E(t)$$



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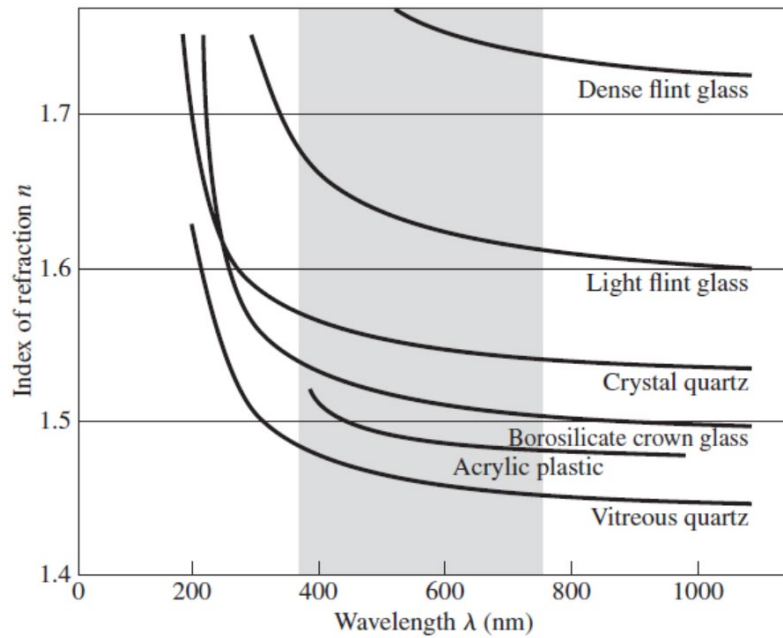
- ❑ Without the driving force, the oscillator will vibrate at its resonant frequency ω_0 ;
- ❑ If $\omega < \omega_0$; $x(t)$ and $E(t)$ have the same sign, meaning that the oscillator can follow the applied force, i.e., is in phase with it.
- ❑ If $\omega > \omega_0$; $x(t)$ and $E(t)$ have the opposite sign, meaning that the displacement $x(t)$ is in a direction opposite to that of the instantaneous force oscillator can follow the applied force $q_e E(t)$, i.e., is 180° out-of-phase with it.

❑ Dispersion equation:

$$P(t) = q_e x(t) N = \frac{q_e^2 N E(t) / m_e}{(\omega_0^2 - \omega^2)}$$

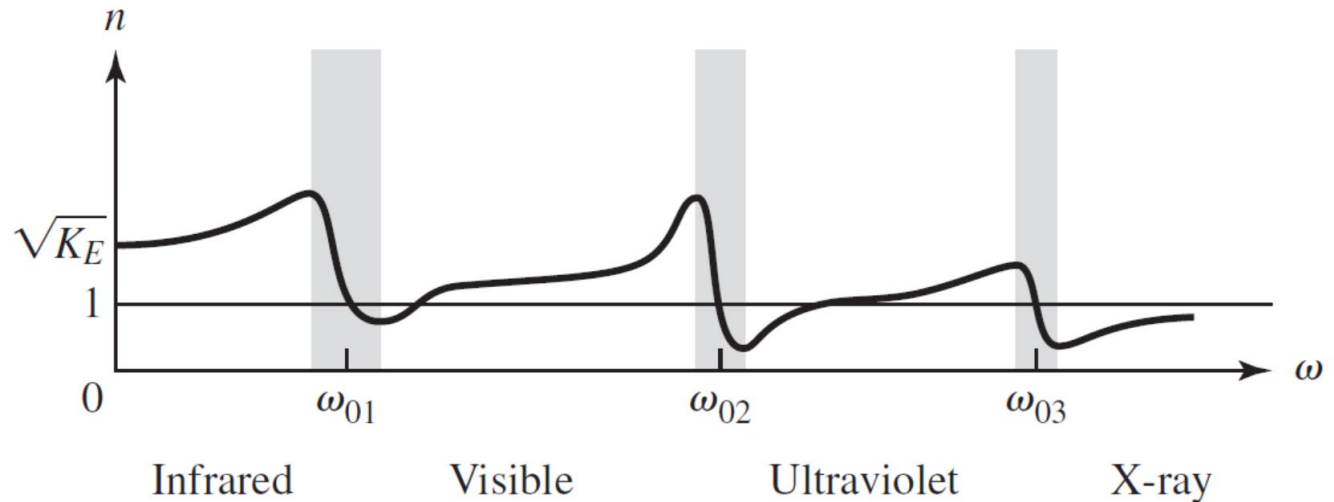
$$\epsilon = \epsilon_0 + \frac{P(t)}{E(t)} = \epsilon_0 + \frac{q_e^2 N / m_e}{(\omega_0^2 - \omega^2)}$$

$$n^2(\omega) = 1 + \frac{N q_e^2}{\epsilon_0 m_e} \left(\frac{1}{\omega_0^2 - \omega^2} \right)$$



$$n^2(\omega) = 1 + \frac{Nq_e^2}{\epsilon_0 m_e} \sum_j \left(\frac{f_j}{\omega_{0j}^2 - \omega^2} \right)$$

- ☐ Normal dispersion
- ☐ Anomalous dispersion



Homework

□ 3.1, 3.2, 3.6 , 3.20

3.1 Consider the plane electromagnetic wave (in SI units) given by the expressions $E_x = 0$, $E_y = 2 \cos[2\pi \times 10^{14}(t - x/c) + \pi/2]$, and $E_z = 0$.

- (a) What are the frequency, wavelength, direction of motion, amplitude, initial phase angle, and polarization of the wave?
- (b) Write an expression for the magnetic flux density.

3.2 Write an expression for the $\vec{E}$ - and $\vec{B}$ -fields that constitute a plane harmonic wave traveling in the $+z$ -direction. The wave is linearly polarized with its plane of vibration at 45° to the yz -plane.

3.6 The electric field of an electromagnetic wave traveling in the positive x -direction is given by

$$\vec{E} = E_0 \hat{j} \sin \frac{\pi z}{z_0} \cos(kx - \omega t)$$

- (a) Describe the field verbally. (b) Determine an expression for k . (c) Find the phase speed of the wave.

3.20 A 1.0-mW laser has a beam diameter of 2 mm. Assuming the divergence of the beam to be negligible, compute its energy density in the vicinity of the laser.